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# The Interaction Tax: How Communication Erases Diversity in Multi-Agent Teams

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## Abstract

1 Multi-agent LLM systems build on the assumption that agent collaborations can  
2 outperform single agents. Interestingly, recent work shows single agents can  
3 match multi-agent systems under equal budgets [Tran and Kiela, 2026, Xu et al.,  
4 2026, Li et al., 2024, Jarrett et al., 2025]. In this work, we further study the true  
5 reasons behind this behavior, by looking deeper into how and when agent interac-  
6 tions help. We find that when different agents read each other’s work, they reach  
7 a consensus immediately, erasing the diversity that motivated using different mod-  
8 els. We call this the interaction tax. This interaction tax depends on the task struc-  
9 ture. For open-ended tasks, the first round of interaction makes solutions worse  
10 57% of the time, because agents converge to one final answer instead of explor-  
11 ing different approaches. Letting agents independently work on solutions first can  
12 mitigate this effect. On tasks with clear rules, e.g. Knapsack, critique helps agents  
13 fix specific errors and raises success rate from 0% to 47–73% ( $p<0.003$ ). These  
14 findings suggest a design principle for multi-agent systems: interaction should be  
15 used selectively, preserving independent exploration for open-ended problems and  
16 applying critique only when errors are concrete and verifiable.

## 17 1 Introduction

18 Scientific optimization rewards the single best solution found. Different model families approach  
19 open problems with different inductive biases, so a diverse team should cover more of the search  
20 space than any single model. The same logic holds for teams of human scientists, where the benefit  
21 of multiple perspectives fades once everyone begins reasoning from the same intermediate draft.  
22 In that setting the wisdom-of-crowds effect [Surowiecki, 2004] weakens because correlated errors  
23 compound rather than cancel [Lorenz et al., 2011]. We ask whether the same pattern appears in  
24 multi-agent AI systems.

25 In practice, different model families produce structurally different solutions. In our tasks, Claude  
26 solves Circle Packing perfectly (normalized quality score  $Q=1.0$ ) while GPT-4o scores 0.519, but  
27 on Erdős Overlap GPT-4o leads ( $Q=0.710$ ) while Gemini scores zero. This complementarity has  
28 motivated multi-agent frameworks across coding [Hong et al., 2023, Wu et al., 2023], mathematical  
29 reasoning [Du et al., 2024, Chen et al., 2025], and scientific discovery [Romera-Paredes et al., 2024,  
30 Georgiev et al., 2025], hoping it leads to better joint solutions.

31 That premise holds only if diversity survives the interaction step meant to exploit it (Figure 1).  
32 Recent evidence suggests that it often does not. Under equal token budgets, single models match or  
33 exceed multi-agent systems [Tran and Kiela, 2026, Xu et al., 2026]. Scaling independent samples via  
34 voting improves performance without any interaction [Li et al., 2024]. Self-organizing LLM teams  
35 can even underperform their best member, losing up to 37.6% through “integrative compromise”  
36 [Jarrett et al., 2025]. However, these studies either use same-model teams or do not control for

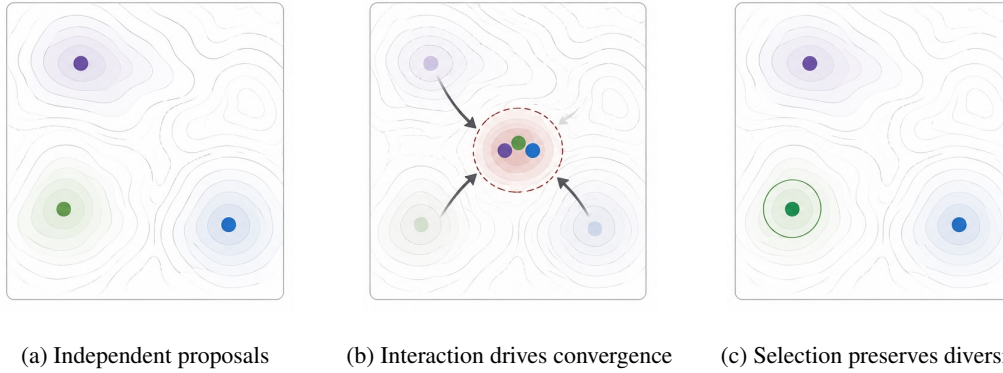


Figure 1: **The interaction tax on optimization tasks.** Different model families (Claude, GPT-4o, Gemini) find structurally different solutions. When agents interact, their proposals converge and diversity is lost. When agents generate independently and the best is selected, diversity is preserved.

37 model diversity, so it remains unclear whether the loss comes from interaction itself or from the  
 38 absence of genuine diversity across model families.

39 To separate these factors, we independently vary backbone diversity and interaction protocol across  
 40 11 tasks (9 optimization, 2 constraint satisfaction) with 10 configurations spanning single-agent  
 41 baselines and multi-agent architectures. To avoid attributing sampling gains to collaboration, we  
 42 match single-model sample counts to multi-agent budgets. Optimization comparisons are scored on  
 43 hidden evaluators that agents never query during search, which reduces the chance of overfitting to  
 44 the visible scorer. We use two summary metrics. Marginal Epistemic Gain (MEG) asks whether  
 45 a configuration beats every matched single-agent baseline. Marginal Interaction Gain (MIG) asks  
 46 whether interaction improves on running the same agents independently under the same sample  
 47 budget.

48 We call this pattern an *interaction tax*. Interaction helps when agents are mostly refining a shared  
 49 direction, but once the team is genuinely diverse it often does the opposite. After a single round  
 50 of full-output exchange, proposals move toward one another, and synthesis usually returns one pro-  
 51 posal with little real combination. We see the same thing within a single backbone—iterative meth-  
 52 ods lose to independent sampling on 4 of 5 optimization tasks, with the fifth essentially tied. But  
 53 on tasks with trivially checkable violations, critique can repair specific errors. When verification  
 54 itself requires reasoning, the critique step is more likely to collapse diversity than to improve the an-  
 55 swer. A crossover protocol that forces component-level combination reduces copying and improves  
 56 performance on 4 of 5 tasks.

57 **Contributions.** Compared with prior equal-budget comparisons, three differences matter here. The  
 58 collapse we observe is fast, unlike the gradual convergence reported in reasoning debates [Liang  
 59 et al., 2023]—it often happens after the first full-solution exchange. We also track diversity at  
 60 the level of final solutions rather than text alone, using task-specific metrics (Appendix A). On  
 61 average, pairwise *solution* distance falls from 0.48 to 0.34 after interaction. And while Jarrett et al.  
 62 [2025] describe multi-agent losses as “integrative compromise,” our benchmark usually shows a  
 63 narrower failure mode. The synthesizer picks one proposal and drops the rest. Architectures that  
 64 keep proposers separate, such as Mixture of Agents [Wang et al., 2025], are much less exposed to  
 65 this pattern.

66 Relative to the workshop version, the extended experiments make the picture more concrete.  
 67 MAGICoRe-Gemini loses to BoN-Gemini on 4 of 5 optimization tasks ( $d > 1.2$ ,  $p < 0.01$ ), so  
 68 the tax is not limited to mixed backbones. On Knapsack, critique in the presence of checkable  
 69 violations increases solution diversity 3-fold across seeds, whereas on tasks that rely on holistic  
 70 judgment it reduces diversity (ratio 0.74). Crossover synthesis also lowers the copying rate from  
 71  $\sim 80\%$  to  $\sim 36\%$  while outperforming standard synthesis on 4 of 5 tasks.

Table 1: Benchmark tasks. Nine optimization tasks and two constraint satisfaction tasks.

| Task             | Domain        | Type    | Objective                                   |
|------------------|---------------|---------|---------------------------------------------|
| Circle Packing   | Geometry      | Opt.    | Max radius of $n$ circles in unit square    |
| Difference Bases | Combinatorics | Opt.    | Cover all residues mod $n$                  |
| Erdős Overlap    | Analysis      | Opt.    | Min $\max_t \int  f(x)f(x+t)  dx$           |
| Flat Polynomials | Algebra       | Opt.    | Min $\ p\ _\infty$ for $\pm 1$ coefficients |
| TSP-50 / TSP-100 | Routing       | Opt.    | Minimize tour length                        |
| MaxCut           | Graph theory  | Opt.    | Maximize cut value                          |
| LJ- $n=41$       | Chemistry     | Opt.    | Minimize Lennard-Jones potential            |
| Molecule QED     | Drug design   | Opt.    | Maximize drug-likeness (QED)                |
| Knapsack-50      | Combinatorics | Constr. | Max value s.t. weight limit                 |
| 3AP-Free-100     | Combinatorics | Constr. | Largest 3-AP-free subset                    |

## 2 Related Work

**Single vs. multi-agent comparisons.** Single-agent “thinking” matches multi-agent debate under equal token budgets [Tran and Kiela, 2026], a simple RAG baseline matches or exceeds multi-agent workflows [Xu et al., 2026], scaling independent samples via majority voting improves accuracy without interaction [Li et al., 2024], and self-organizing teams lose up to 37.6% through “integrative compromise” that averages expert and non-expert views [Jarrett et al., 2025]. All four studies use same-model teams or do not control for backbone diversity. Our work isolates backbone diversity as a variable, introduces a hidden-evaluator split, and identifies an error-verifiability boundary that determines when the tax reverses.

**Iterative refinement.** Self-Refine [Madaan et al., 2023] and Reflexion [Shinn et al., 2023] show that iterative self-critique improves generation quality. MAgICoRe [Chen et al., 2025] extends this with a multi-agent coarse-to-fine pipeline. Tree-of-Thoughts [Yao et al., 2024] explores multiple reasoning paths with backtracking. We find that the benefits of iteration are task-dependent. On constraint tasks with verifiable errors, external critique is the active ingredient (Self-Refine  $\approx$  Best-of- $N$ , while MAgICoRe significantly outperforms both). On optimization tasks, iteration within a single backbone hurts on 4 of 5 comparisons.

**Evolutionary approaches.** FunSearch [Romera-Paredes et al., 2024], Evolution through Large Models [Lehman et al., 2023], and AlphaEvolve [Georgiev et al., 2025] embed LLMs inside evolutionary loops for mathematical discovery. These approaches preserve population diversity through selection pressure rather than through agent interaction, avoiding the convergence we study. Our crossover protocol is inspired by genetic algorithms and operates on complete solutions.

**Ensemble diversity theory.** The ensemble diversity decomposition [Krogh and Vedelsby, 1994, Hansen and Salamon, 1990, Lakshminarayanan et al., 2017] shows that ensemble error equals mean member error minus a diversity term. When members communicate and converge, diversity shrinks toward zero. Our results fit that picture: communication increases inter-agent correlation and reduces the diversity term.

## 3 Experimental Setup

**Tasks.** We test 11 tasks spanning mathematical optimization and constraint satisfaction (Table 1). Nine are optimization tasks where quality is a continuous score with no single correct answer (e.g., minimize tour length, maximize packing density). Two are constraint satisfaction tasks where a solution either satisfies all rules or does not (e.g., does the knapsack exceed its weight limit?). Four optimization tasks are adapted from the AlphaEvolve suite [Georgiev et al., 2025] with smaller parameterizations. Raw scores are normalized to  $Q \in [0, 1]$ , clipped, where 0 is the trivial baseline and 1 is the best known result. Circle Packing remains an optimization task in the main benchmark. Later feasibility analyses use a derived valid-packing view to isolate whether critique helps repair explicit violations.

**Hidden evaluator.** Each task has a visible evaluator (queryable during search) and a hidden evaluator (run once on the final solution). The hidden evaluator prevents overfitting to the visible scorer.

Table 2: Ten configurations used in the main experiments. BoN = Best-of- $N$ .

| Configuration | Type     | Mechanism                                                                  |
|---------------|----------|----------------------------------------------------------------------------|
| Single-Shot   | Baseline | One model, one attempt                                                     |
| BoN ( $N=8$ ) | Baseline | 8 independent samples, return best                                         |
| Self-Refine   | Baseline | Generate, critique, revise $\times 3$ [Madaan et al., 2023]                |
| VGS           | Baseline | Evolutionary loop, visible scores as fitness [Romera-Paredes et al., 2024] |
| Chain         | Multi    | 3–4 agents in sequence, each revises prior                                 |
| MAGICoRe      | Multi    | Solver→reviewer→refiner, 2 rounds [Chen et al., 2025]                      |
| Debate        | Multi    | 2 agents, 2 critique rounds + synthesizer [Du et al., 2024]                |
| HPE           | Multi    | Planner + executors + integrator [Wang et al., 2023]                       |
| MoA           | Multi    | 3 proposers (independent) + 1 synthesizer [Wang et al., 2025]              |
| Crossover     | Multi    | 3 proposers + component-level synthesis                                    |

Table 3: Best-of- $N$  Q-scores per backbone on six tasks where scores clearly diverge. Bold = best backbone. Each model family wins on different tasks.

| Task           | Claude       | GPT-4o       | Gemini       | Winner |
|----------------|--------------|--------------|--------------|--------|
| Circle Packing | <b>1.000</b> | 0.519        | 0.745        | Claude |
| TSP-50         | <b>0.141</b> | —            | 0.115        | Claude |
| Erdős          | 0.131        | <b>0.710</b> | 0.000        | GPT-4o |
| TSP-100        | 0.013        | <b>0.021</b> | 0.010        | GPT-4o |
| DiffBases      | 0.209        | 0.264        | <b>0.610</b> | Gemini |
| Flat Poly      | 0.000        | 0.007        | <b>0.143</b> | Gemini |

Visible and hidden rankings diverge on three of nine optimization tasks (Appendix K.1). To check that the visible evaluator tracks the hidden one, we run a K-curve analysis on MaxCut, varying the number of evaluator calls from 1 to 25. Both dev and hidden scores generally increase with  $K$ , and no systematic proxy overfit is detected (Appendix L.5).

**Configurations.** We test ten configurations (from a larger pool of 48, see Appendix D) built from Claude Sonnet 4 [Anthropic, 2025], GPT-4o [OpenAI, 2024], and Gemini 2.5 Flash [Google DeepMind, 2025], summarized in Table 2. MoA is the only standard configuration where proposers never read each other’s outputs. Crossover extends MoA by instructing the synthesizer to combine parts from different proposals. Each configuration runs under identical resource budgets ( $T=200K$  tokens,  $W=600s$  wall time,  $C=30s$  per call,  $K=25$  evaluator calls). Each (task, configuration) cell uses 5–10 seeds.

**Metrics.** We define two metrics to separate the effects of diversity and interaction:

*Marginal Epistemic Gain* (MEG): For configuration  $c$  on task  $t$ ,  $MEG(c, t) = \bar{Q}(c, t) - \max_{b \in \mathcal{B}} \bar{Q}(b, t)$ , where  $\mathcal{B}$  is the set of single-agent baselines matched by backbone. Positive MEG means the configuration outperformed every single-agent baseline.

*Marginal Interaction Gain* (MIG):  $MIG(c, t) = \bar{Q}(c, t) - \bar{Q}(c_{\text{parallel}}, t)$ , where  $c_{\text{parallel}}$  runs the same agents independently under equal budgets. Positive MIG means interaction improved over independent generation. The diverse parallel baseline is already strong because it benefits from coverage, making it a demanding reference point. Formal definitions are in Appendix A.

## 4 Results

### 4.1 Backbone Diversity Creates Coverage

Each model is the top scorer on at least one task (Table 3), and every same-model team catastrophically fails ( $Q=0$ ) on at least one task. The diverse team and the best same-model team score similarly on aggregate ( $Q=0.214$  vs.  $0.221$ ), but the diverse team never catastrophically fails and wins on 5 of 9 tasks. To test whether this coverage translates into a measurable performance advantage, we run a controlled factorial experiment.

Table 4:  $2 \times 2$  factorial estimates ( $N=198$ , 10,000 bootstrap). Diversity is the only factor whose CI excludes zero.

| Term                         | Estimate | 95% CI           |
|------------------------------|----------|------------------|
| Intercept (same + selection) | +0.181   | [+0.146, +0.220] |
| Diversity                    | +0.195   | [+0.125, +0.262] |
| Synthesis                    | +0.044   | [−0.023, +0.113] |
| Diversity $\times$ Synthesis | +0.035   | [−0.103, +0.172] |

Table 5: Mean MIG per configuration family (8 tasks).  $P(\text{same} > \text{div})$ : bootstrap probability that same-model MIG exceeds diverse MIG.

| Configuration | Same MIG | Div MIG | $P$ |
|---------------|----------|---------|-----|
| Chain         | +0.051   | −0.024  | 97% |
| MAGICoRe      | +0.044   | −0.035  | 99% |
| Debate        | +0.012   | −0.078  | 88% |
| HPE           | −0.163   | −0.145  | 38% |
| MoA           | +0.012   | +0.016  | 42% |

## 136 4.2 The Diversity Advantage: A Controlled Factorial Test

137 A  $2 \times 2$  factorial crosses backbone diversity (same vs. diverse proposers) with synthesis (MoA  
 138 synthesis vs. best-score selection) on Erdős, DiffBases, and MolQED, the three tasks with suffi-  
 139 cient Q-score variance ( $N=198$  observations, task-stratified bootstrap). The diversity coefficient is  
 140  $\beta_{\text{div}} = +0.195$  (CI [+0.125, +0.262]), positive in all 10,000 bootstrap resamples (Table 4). The  
 141 synthesis coefficient is +0.044 (CI straddles zero). Leave-one-out analysis (Appendix H.1) shows  
 142 the finding is partially driven by Erdős. Removing it drops the coefficient to +0.052 (CI crosses  
 143 zero), while removing either other task leaves it significant.

144 **Robustness across synthesizers.** The  $2 \times 2$  factorial uses Claude as the synthesizer. To rule out  
 145 synthesizer-identity confounds, we replicate with GPT-4o and Gemini as synthesizers (Appendix I).  
 146 The diversity coefficient holds up for all three. Claude  $\beta = +0.234$  ( $p < 0.001$ ), GPT-4o  $\beta = +0.176$   
 147 ( $p=0.007$ ), Gemini  $\beta = +0.170$  ( $p=0.010$ ). But does interaction preserve it?

## 148 4.3 The Interaction Tax

149 Same-model interaction can refine proposals. Chain, MAGICoRe, and Debate all produce positive  
 150 same-model MIG (Table 5). With diverse backbones, all three turn negative (Chain −0.024, MAGI-  
 151 CoRe −0.035, Debate −0.078), with bootstrap  $P(\text{same} > \text{div}) \geq 88\%$ . For MAGICoRe and Debate,  
 152 only the models change across conditions, so the loss comes from the diversity that interaction erases.  
 153 MoA’s MIG remains positive from same-model to diverse (+0.012  $\rightarrow$  +0.016), showing that the  
 154 damage comes from the interaction step itself. No configuration achieves positive aggregate MEG  
 155 (Figure 2).

## 156 4.4 How Convergence Happens

157 **Immediate diversity collapse.** On Erdős, diverse Debate achieves a strong intermediate score at  
 158 round 2 (when agents exchange only critiques) but regresses at round 3 once agents read each other’s  
 159 full solutions (trajectories in Appendix J). Across the three tasks analyzed (Table 19), mean pairwise  
 160 distance between agents’ solution representations falls from 0.48 (MoA, no synthesis) to 0.34 after  
 161 synthesis, matching same-model MoA levels.

162 Text-level similarity shows the same pattern. Diverse proposers produce outputs with cosine similar-  
 163 ity  $\sim 0.43$ , versus  $\sim 0.74$  for same-model Best-of- $N$  (Appendix J.3). Debate pushes this higher over  
 164 rounds. Same-model debate rises from 0.646 to 0.717, while diverse debate starts at 0.728, already  
 165 converged after reading each other’s work.

166 MAGICoRe also makes solutions across seeds more similar. On Circle Packing, inter-seed pairwise  
 167 distance under MAGICoRe is only  $0.74 \times$  the distance under Best-of- $N$  (Appendix J.4).

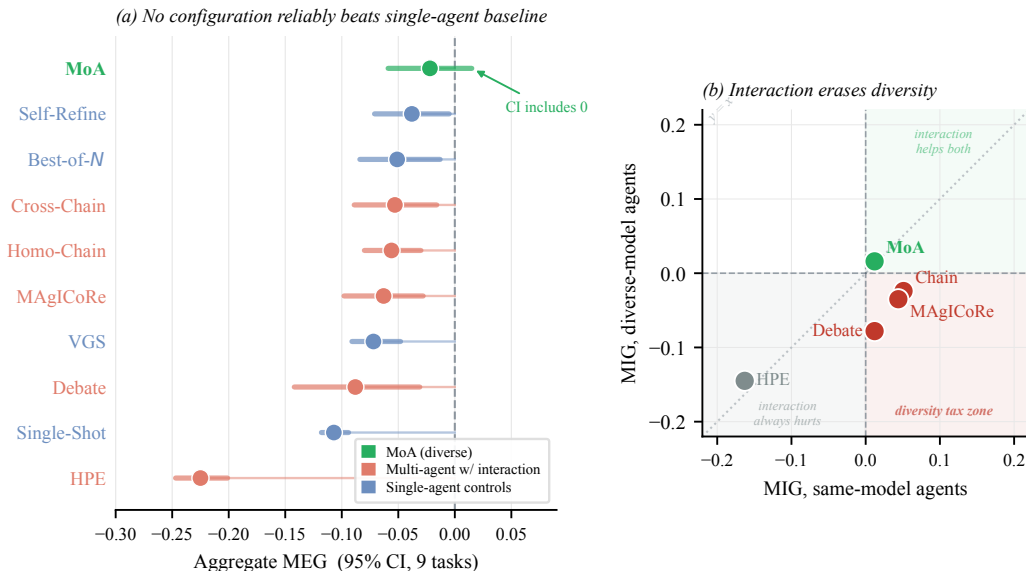


Figure 2: (a) Aggregate MEG for all ten configurations (95% CI, 9 tasks). MoA is the only configuration whose confidence interval includes zero. (b) MIG quadrant scatter ( $x$ -axis = same-model MIG,  $y$ -axis = diverse-model MIG). MagICoRe, Chain, and Debate occupy the *interaction tax zone* (bottom-right quadrant), where interaction helps same-model agents but hurts once backbones are diverse. MoA escapes to the top-right because its proposers never see each other’s outputs.

Table 6: Synthesis output classified relative to the best-scoring proposer (MoA, diverse backbones,  $n=10$ ).

| Task           | Improved | Copied | Degraded |
|----------------|----------|--------|----------|
| Circle Packing | 60%      | 40%    | 0%       |
| Molecule QED   | 20%      | 80%    | 0%       |
| Erdős          | 10%      | 80%    | 10%      |
| Flat Poly      | 0%       | 100%   | 0%       |
| TSP-100        | 0%       | 100%   | 0%       |
| TSP-50         | 0%       | 90%    | 10%      |
| DiffBases      | 7%       | 43%    | 50%      |

168 **Synthesis copies instead of combining.** The problem is that synthesis rarely combines proposals  
169 in any substantive way. On five of seven tasks, MoA synthesis copies the best proposer’s output  
170  $\geq 80\%$  of the time (Table 6). On Erdős, GPT-4o outputs the identical trivial constant on 100% of  
171 seeds ( $Q=0.710$ ), so any synthesis step converges to that single point regardless of what Claude or  
172 Gemini proposed.

173 Jarrett et al. [2025] identify “integrative compromise” where teams average expert and non-expert  
174 views. In our setting the failure mode is simpler. The synthesizer reproduces one agent’s solution  
175 nearly unchanged, collapsing the team’s output to a single point in solution space.

#### 176 4.5 The Tax Operates Within a Single Backbone

177 The results so far involve mixed backbones. To check whether interaction alone causes the tax, we  
178 compare iterative methods against Best-of- $N$  using only the Gemini backbone, removing diversity  
179 as a variable entirely ( $n=15$  per condition).

180 MagICoRe-Gemini loses to BoN-Gemini on 4 of 5 optimization tasks, with large effect sizes ( $d >$   
181 1.2,  $p < 0.01$  on all four, Table 7). MolQED shows no significant difference ( $p=0.93$ ). The single  
182 win appears in the Circle Packing feasibility view: because valid packings must satisfy a hard non-  
183 overlap constraint, feasibility is a useful diagnostic alongside the optimization score. Under that

Table 7: Within-backbone comparison: MAgICoRe-Gemini vs. BoN-Gemini ( $n=15$ ). Positive  $d$  means iterative refinement is worse on the optimization rows; the Circle Packing row reports feasibility only. The interaction tax operates even without diverse backbones.

| Task                   | MAgICoRe   | BoN           | Cohen’s $d$ | $p$     |
|------------------------|------------|---------------|-------------|---------|
| Circle Packing (feas.) | <b>80%</b> | 27%           | —           | 0.005   |
| Flat Poly (raw)        | 2.087      | <b>1.899</b>  | +1.25       | 0.002   |
| TSP-100 (raw)          | 53,351     | <b>50,786</b> | +1.42       | 0.001   |
| TSP-50 (raw)           | 24,167     | <b>23,103</b> | +1.23       | 0.002   |
| Erdős (raw)            | MAgI worse |               | +1.91       | < 0.001 |
| MolQED (raw)           | 0.863      | 0.859         | +0.05       | 0.93    |

Table 8: Constraint feasibility rates by backbone condition ( $n=10$  per cell). On Knapsack (trivially checkable), diverse backbones help. On 3AP-Free (reasoning-required), diverse backbones hurt.

| Task         | Config   | Same | Diverse      | Direction  |
|--------------|----------|------|--------------|------------|
| Knapsack-50  | MAgICoRe | 1/10 | 2/10         | $\approx$  |
|              | Debate   | 2/10 | <b>10/10</b> | Div. helps |
| 3AP-Free-100 | MAgICoRe | 6/10 | 3/10         | Div. hurts |
|              | Debate   | 6/10 | <b>0/10</b>  | Div. hurts |

view, MAgICoRe achieves 80% feasibility versus BoN’s 27% ( $p=0.005$ ). The pattern replicates on the GPT-4o backbone (Appendix L.2).

On open-ended optimization tasks, independent sampling outperforms iterative refinement regardless of backbone diversity, because each sample explores a different region of solution space.

**Self-refine vs. external critique.** On tasks where we evaluate explicit validity, the extra gain comes from external critique rather than from iteration by itself. Self-Refine performs comparably to Best-of- $N$  on the Circle Packing feasibility view (27% vs. 27% valid packings) and only marginally better on Knapsack (13% vs. 0%). MAgICoRe beats both by a wide margin, reaching 80% on Circle Packing validity ( $p=0.005$  vs. BoN) and 47% on Knapsack ( $p=0.003$ ). In these settings, the multi-agent critique step matters more than self-critique alone (Appendix L.3). Why does validity repair help while open-ended optimization does not?

#### 4.6 The Verifiability Boundary

What matters here is whether feedback can point to a concrete, trivially checkable mistake (Table 8).

On Knapsack-50, where a weight violation is an arithmetic check, diverse Debate achieves 10/10 feasibility versus 2/10 for same-model Debate. The tax reverses because agents can identify and fix concrete errors. In additional experiments on the composable-2x2 dataset, diverse MoA achieves a +940 score advantage over same-model MoA on Knapsack, with 100% crossover feasibility (Appendix N.4). On 3AP-Free-100, where verifying a three-term arithmetic progression requires reasoning, diverse Debate drops to 0/10 versus 6/10 for same-model. The pattern replicates across MAgICoRe. On this task, agents generate plausible-sounding but incorrect critiques that mislead their partners. On optimization tasks, the first critique round degrades solutions 57% of the time (17/30 runs across two tasks), while on constraint tasks with checkable violations it never does (0/30,  $p < 0.001$  Fisher exact, Appendix G). The difference is whether feedback can point to a specific error or only offer a holistic judgment.

#### 4.7 Why the Boundary Exists

The contrast extends beyond performance. Critique changes solution *diversity itself*, and it does so differently depending on whether errors are checkable. We measure inter-seed pairwise solution distance under MAgICoRe versus Best-of- $N$  using task-specific metrics (Appendix A, Table 9).

On Knapsack, MAgICoRe produces solutions that are  $3\times$  more diverse across seeds than Best-of- $N$ . When critique identifies a specific constraint violation (“this item exceeds the weight limit”),

Table 9: Ratio of inter-seed solution distance (MAgICoRe / BoN). Ratio  $> 1$  means critique *increased* diversity. On Knapsack (checkable errors), critique drives agents to explore different constraint-satisfying solutions. On the Circle Packing feasibility view and on open-ended optimization tasks, critique instead narrows the solution set.

| Task                   | Type          | MAgI dist | BoN dist | Ratio       |
|------------------------|---------------|-----------|----------|-------------|
| Knapsack-50            | Constraint    | 0.495     | 0.162    | <b>3.06</b> |
| Circle Packing (feas.) | Validity view | 1.526     | 2.069    | 0.74        |
| Flat Poly              | Optimization  | 0.907     | 0.966    | 0.94        |
| TSP-100                | Optimization  | 0.974     | 0.985    | 0.99        |

Table 10: Crossover vs. MoA vs. MoA-nosynth (hidden scores,  $n=10$ ). Crossover wins 4/5 tasks against MoA and 5/5 against MoA-nosynth. Lower is better except Knapsack (higher is better).

| Task      | Crossover     | MoA          | MoA-nosynth | Winner    |
|-----------|---------------|--------------|-------------|-----------|
| DiffBases | <b>10.99</b>  | 25.70        | 23.51       | Crossover |
| Erdős     | 0.298         | <b>0.250</b> | 0.348       | MoA       |
| Knapsack  | <b>842.2</b>  | 284.8        | 761.3       | Crossover |
| Set-Cover | <b>491</b>    | 100,343      | 200,309     | Crossover |
| TSP-50    | <b>22,644</b> | 23,217       | 23,173      | Crossover |

different seeds can repair it in different ways, producing a wider set of feasible solutions. On the Circle Packing feasibility view, where checking validity still requires geometric reasoning about overlap, critique instead pulls solutions toward a narrower family (ratio 0.74). On optimization tasks without hard constraints, critique has only a modest effect on diversity (ratios near 1.0).

When an error is checkable, there are many ways to fix it, so different seeds diverge. By contrast, when feedback is holistic (“make it better”), seeds tend to move toward the same perceived optimum. Given that synthesis defaults to selecting one proposal, we next test whether explicitly forcing combination helps.

#### 4.8 Crossover Synthesis: A Mitigation

Standard MoA synthesis copies the best proposer  $\geq 80\%$  of the time (Section 4.4). Crossover synthesis instructs the synthesizer to combine *components* from different proposals, for example taking the item selection strategy from one knapsack solution and the pruning heuristic from another. We compare crossover against standard MoA and against MoA without a synthesis step (MoA-nosynth, where the best-scoring independent proposal is returned directly) on five tasks with 10 seeds each (Table 10).

**Reduced copying.** The average copying rate drops from  $\sim 80\%$  (standard MoA) to  $\sim 36\%$  (crossover). Lower copying tracks with better scores across all four tasks where crossover wins.

**Improved constraint feasibility.** On Knapsack, crossover achieves 100% feasibility (10/10) versus 70% for standard MoA and 90% for MoA-nosynth. By combining constraint-satisfying components from different proposals, crossover avoids introducing new violations.

Crossover loses to standard MoA only on Erdős, where the best solution is a trivial constant that cannot be meaningfully split into components. On all other tasks, component-level combination outperforms copying a single proposal.

## 5 Discussion and Conclusion

Our experiments show that interaction destroys the diversity it is meant to use. Different model families cover different solution regions and prevent catastrophic failure. However, when agents exchange full outputs, proposals converge within a single critique round, with synthesis copying the best proposer rather than recombining components. MoA, where proposers never see each other’s outputs, is the only configuration whose diverse MIG remains positive. This reverses only when errors are checkable.

The diversity advantage ( $\beta = +0.195$ ) is strongest on tasks where models make structurally different errors, so between-model variance exceeds within-model variance (Appendix M.2). On tasks where all models approach the problem similarly (TSP-100, MaxCut), diversity adds little. In the harder external-2x2 experiments, diversity helps on only 1 of 6 tasks while synthesis helps on 5 of 6, suggesting that once most proposals are poor or infeasible, the bottleneck shifts from coverage to correction (Appendix N.6). Doubling the compute budget improves 12 of 15 task-protocol pairs, but the gains remain modest (1–38%, Appendix N.3). Model diversity often looks like a better lever than spending more compute.

Our results suggest the first question for any multi-agent system is whether agents should interact at all. Agents could share only their scores rather than full solutions, reducing convergence pressure. Synthesis could combine parts from different proposals rather than copying one wholesale. Once one proposal becomes the anchor, the rest of the discussion tends to orbit it, making the communication channel itself a design choice.

**Limitations.** All tasks are mathematical or combinatorial optimization, so the interaction tax may not transfer to natural-language generation, code synthesis, or other settings where “diversity” has a different meaning. The coverage finding is partially driven by Erdős (removing it drops  $\beta_{\text{div}}$  to +0.052 with CI crossing zero), though the multi-synthesizer replication confirms the direction is robust across all three synthesizers. Five seeds per condition is marginal for per-task claims, and the verifiability boundary rests on two constraint tasks in the main benchmark plus three in extended experiments. We test three commercial model families. Open-source models may exhibit different convergence dynamics, though preliminary evidence from HP-Folding suggests open-source diversity can also help (MAgICoRe-mixed-OSS achieves 4/5 valid folds versus 0/5 for same-model, Appendix N.7). The crossover results are limited to 5 tasks with a single synthesizer and deserve further study. Code and data are available at <https://anonymous.4open.science/r/interaction-tax-3244>.

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## 316 A Formal Metric Definitions

317 Raw scores are normalized to  $Q(s) = (s - b)/(r - b)$ , clipped to  $[0, 1]$ , where  $b$  is the trivial baseline  
318 and  $r$  the best known result.

319 **MEG.**  $\text{MEG}(c, t) = \bar{Q}(c, t) - \max_{b \in \mathcal{B}} \bar{Q}(b, t)$ , where  $\mathcal{B}$  is the set of single-agent baselines sharing  
320 the same backbone(s) as  $c$ , with sample count matched to  $c$ ’s budget.

321 **MIG.**  $\text{MIG}(c, t) = \bar{Q}(c, t) - \bar{Q}(c_{\text{par}}, t)$ , where  $c_{\text{par}}$  runs the same agent(s) independently under the  
322 same total budget. For diverse configurations,  $c_{\text{par}}$  is MoA-nosynth (three proposers, best selected  
323 by visible score). For same-model configurations,  $c_{\text{par}}$  is Best-of- $N$  with  $N$  matched to the total call  
324 count.

325 **Pairwise solution distance.** To measure structural diversity between solutions, we compute pair-  
326 wise distances using task-appropriate metrics on the final solution representations. For tasks  
327 with binary selection vectors (Knapsack), we use normalized Hamming distance (fraction of dif-  
328 fering items). For tasks with coefficient vectors (Flat Polynomials), we use cosine distance  
329 ( $1 - \text{cosine similarity}$ ). For tasks producing tours (TSP), we convert each tour to its edge set and com-  
330 pute Jaccard distance ( $1 - |A \cap B|/|A \cup B|$ ). For tasks with coordinate outputs (Circle Packing), we  
331 compute L2 distance between flattened position vectors. For function-valued tasks (Erdős Overlap),  
332 we interpolate the discretized function values to a common grid of 200 points and compute cosine  
333 distance. For tasks with integer-set outputs (DiffBases), we use Jaccard distance on the solution sets.  
334 For tasks with string representations (Molecule QED), we use normalized Levenshtein distance on  
335 SMILES strings. All metrics are bounded in  $[0, 1]$ . Inter-seed distances (Section 4.7, Table 9) are  
336 means over all  $\binom{n}{2}$  seed pairs within a condition. The inter-agent distances in Section 4.4 (Table 19)  
337 use the same task-specific metrics applied to solutions from different agents within a single run,  
338 averaged across three tasks.

## 339 B Workflow Recipe

340 One way to use the paper’s findings in practice is the following workflow for AI-assisted optimiza-  
341 tion:

- 342 1. Choose 2–3 diverse model families.
- 343 2. Generate candidate solutions independently with no cross-agent visibility.
- 344 3. Rank candidates with a visible deterministic checker.
- 345 4. If the task exposes local, checkable violations, allow critique-and-revise passes. Otherwise
- 346 skip interaction and select among the independent candidates.
- 347 5. Verify the chosen artifact with a stricter held-out evaluator.

## 348 C Reproducibility and Verification

349 The anonymized artifact includes exact prompt templates, pinned model identifiers, task instances,  
350 deterministic evaluators, saved JSON run traces with raw model outputs, and scripts that reproduce  
351 every table and figure from cached results. The artifact runs on Node.js  $\geq 20$  and Python  $\geq 3.10$ ;  
352 no GPU is required. Verification prints a table of recorded vs. recomputed scores; all rows match  
353 exactly. Re-running one task with three seeds takes  $\sim 5$  minutes and costs under \$1 via OpenRouter.

## 354 D Agent Configurations

Table 11: Agent configurations. All operate under the same budget vector ( $T=200K$ ,  $W=600s$ ,  $C=30s$ ,  $K=25$ ).

| Configuration          | Role    | Mechanism                                                     | Ref.                         |
|------------------------|---------|---------------------------------------------------------------|------------------------------|
| Single-Shot            | Control | One model, one attempt                                        | —                            |
| Best-of- $N$ ( $N=8$ ) | Control | 8 independent samples                                         | —                            |
| Self-Refine            | Control | Generate, critique, revise $\times 3$                         | Madaan et al. [2023]         |
| VGS                    | Control | Evolutionary loop (pop. 4, gen. 5)                            | Romera-Paredes et al. [2024] |
| Homo-Chain             | Multi   | 4 same-model agents in sequence                               | This work                    |
| Cross-Chain            | Multi   | 3 diverse agents in sequence                                  | This work                    |
| MAGICoRe               | Multi   | Solver $\rightarrow$ reviewer $\rightarrow$ refiner, 2 rounds | Chen et al. [2025]           |
| Debate                 | Multi   | 2 agents, 2 critique rounds + synth                           | Du et al. [2024]             |
| HPE                    | Multi   | Planner + executors + integrator                              | Wang et al. [2023]           |
| MoA                    | Multi   | 3 proposers + 1 synthesizer                                   | Wang et al. [2025]           |

355 All proposer calls use temperature 0.7. Critique and synthesis calls use 0.3–0.5, consistent across  
356 backbones.

## 357 E Full Per-Task MEG

Table 12: Per-task MEG for all ten configurations. MaxCut and LJ- $n=41$  omitted (all configurations score zero).

| Task           | MoA   | SR    | BoN   | XCh   | HCh   | MAGI  | VGS   | Deb   | SS    | HPE   |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Circle Packing | −0.24 | −0.03 | 0.00  | −0.03 | 0.00  | 0.00  | 0.00  | −0.52 | 0.00  | −1.00 |
| DiffBases      | +0.09 | −0.11 | 0.00  | −0.11 | −0.20 | −0.15 | −0.13 | −0.11 | −0.20 | −0.20 |
| Flat Poly      | +0.01 | −0.06 | −0.08 | −0.08 | −0.08 | −0.08 | 0.00  | −0.04 | −0.08 | −0.08 |
| TSP-100        | −0.02 | −0.02 | −0.01 | −0.01 | −0.02 | −0.01 | 0.00  | −0.02 | −0.01 | −0.02 |
| Erdős          | +0.02 | 0.00  | −0.37 | −0.13 | −0.05 | −0.15 | −0.48 | −0.33 | −0.48 | −0.48 |
| MolQED         | −0.01 | −0.05 | 0.00  | −0.10 | −0.12 | −0.14 | −0.04 | +0.02 | −0.11 | −0.24 |
| TSP-50         | −0.05 | −0.08 | 0.00  | −0.02 | −0.04 | −0.05 | 0.00  | −0.02 | −0.09 | −0.02 |

## 358 F Per-Backbone Coverage and Capability Confound

Table 13: MoA-nosynth aggregate  $Q$ -scores across nine optimization tasks ( $n=10$  per condition). Each same-model team catastrophically fails on at least one task, while the diverse team never does.

| Configuration                      | Aggregate $Q$ | Min task $Q$ |
|------------------------------------|---------------|--------------|
| Diverse (Claude + GPT-4o + Gemini) | 0.214         | > 0          |
| GPT-4o $\times 3$                  | 0.221         | 0.000        |
| Gemini $\times 3$                  | 0.157         | 0.000        |

## 359 G Critique Regression Rates

Table 14: First-round critique regression rate (MAGICoRe, Gemini backbone,  $n=15$  per task).

| Task type              | Regress 0 $\rightarrow$ 1 | Any regress | $n$ |
|------------------------|---------------------------|-------------|-----|
| Constraint (2 tasks)   | <b>0/30</b> (0%)          | 5/30 (17%)  | 30  |
| Optimization (2 tasks) | 17/30 (57%)               | 25/30 (83%) | 30  |

Table 15: Feasibility success rates on two constraint tasks (Gemini backbone,  $n=15$  each). Fisher exact  $p$  compares MAgICoRe to Best-of- $N$ .

| Task         | BoN | Self-Ref | MAgICoRe | $p$    |
|--------------|-----|----------|----------|--------|
| Knapsack-50  | 0%  | 13%      | 47%      | 0.003  |
| 3AP-Free-100 | 0%  | 7%       | 73%      | <0.001 |

## H 2×2 Factorial Details

### H.1 Leave-One-Out Sensitivity

Table 16: Leave-one-out sensitivity of the diversity main effect.

| Task withheld        | $\hat{\beta}_{\text{div}}$ | 95% CI           | $p$    |
|----------------------|----------------------------|------------------|--------|
| None ( $N=198$ )     | +0.195                     | [+0.125, +0.262] | <0.001 |
| Erdős ( $n=158$ )    | +0.052                     | [−0.013, +0.117] | 0.10   |
| DiffBases ( $n=80$ ) | +0.254                     | [+0.144, +0.358] | <0.001 |
| MolQED ( $n=158$ )   | +0.181                     | [+0.097, +0.266] | <0.001 |

### H.2 Per-Task 2×2 Cell Means

Table 17: Mean  $Q$ -scores for each arm of the 2×2 factorial, by task.

| Task      | Same+Sel | Same+Syn | Div+Sel | Div+Syn | Div contr. | Syn contr. |
|-----------|----------|----------|---------|---------|------------|------------|
| DiffBases | 0.202    | 0.071    | 0.237   | 0.287   | +0.126     | −0.041     |
| Erdős     | 0.031    | 0.071    | 0.568   | 0.497   | +0.482     | −0.016     |
| MolQED    | 0.314    | 0.377    | 0.307   | 0.437   | +0.026     | +0.097     |

## I Cross-Synthesizer Replication

Table 18: Diversity coefficient under three synthesizer backbones. The coefficient remains positive and significant for all three.

| Synthesizer      | $\hat{\beta}_{\text{div}}$ | 95% CI           | $p$    |
|------------------|----------------------------|------------------|--------|
| Claude Sonnet 4  | +0.234                     | [+0.106, +0.355] | <0.001 |
| GPT-4o           | +0.176                     | [+0.049, +0.303] | 0.007  |
| Gemini 2.5 Flash | +0.170                     | [+0.044, +0.298] | 0.010  |

Per-task breakdown: On DiffBases, diversity helps with all three synthesizers (+0.14 to +0.22). On Erdős, diversity always helps (+0.37 to +0.50). On MolQED, diversity helps with Claude (+0.06) but slightly hurts with GPT-4o/Gemini (−0.03 to −0.11).

## 367 J Convergence and Diversity Analysis

### 368 J.1 Convergence Mechanism

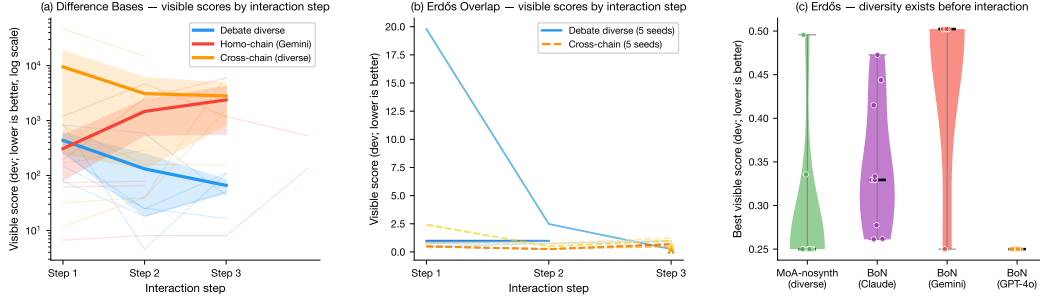


Figure 3: Per-round visible-score trajectories. (a) DiffBases: sequential interaction often improves at step 2 but loses that advantage by the final step. (b) Erdős: diverse interaction improves at round 2 then regresses at round 3 after agents exchange full outputs. (c) Erdős: diverse MoA-nosynth achieves broader coverage than same-backbone Best-of- $N$ .

### 369 J.2 Diversity Collapse

Table 19: Mean pairwise solution distance. After synthesis, diverse MoA distance (0.48) falls to 0.34, matching same-model MoA.

| Configuration             | Avg pairwise distance |
|---------------------------|-----------------------|
| MoA (no synthesis)        | 0.48                  |
| Best-of- $N$ (same model) | 0.40                  |
| Debate                    | 0.38                  |
| MoA (diverse + synthesis) | 0.34                  |
| MoA (same-model)          | 0.34                  |

### 370 J.3 Output Text Similarity

Table 20: Cosine similarity between parallel proposer outputs.

| Condition             | Mean cosine sim | $N$ |
|-----------------------|-----------------|-----|
| MoA diverse proposers | 0.433           | 60  |
| MoA-nosynth proposers | 0.440           | 60  |
| BoN (Claude)          | 0.745           | 30  |
| BoN (Gemini)          | 0.738           | 50  |

371 Diverse proposers produce outputs with cosine similarity  $\sim 0.43$ , versus  $\sim 0.74$  for same-model BoN.  
 372 Same-model debate similarity increases over rounds ( $0.646 \rightarrow 0.717$ ). Diverse debate starts high  
 373 ( $0.728$ )—agents immediately converge.

## 374 J.4 Inter-Seed Diversity Collapse

Table 21: Inter-seed pairwise distance ratio (MAgICoRe / BoN). Ratio  $< 1$  means MAgICoRe makes different seeds more similar. VGS preserves diversity (ratio  $\approx 1.0$ ).

| Task           | MAgICoRe dist | BoN dist | Ratio       | Interp.        |
|----------------|---------------|----------|-------------|----------------|
| Flat Poly      | 0.907         | 0.966    | 0.94        | Less diverse   |
| TSP-100        | 0.974         | 0.985    | 0.99        | $\approx$ Same |
| Circle Packing | 1.532         | 2.075    | <b>0.74</b> | Collapse       |

## 375 K Evaluation Validation

### 376 K.1 Visible–Hidden Rank Agreement

Table 22: Spearman rank correlation between visible and hidden configuration rankings ( $n=10$  configs).

| Task           | $\rho$ | Top-3 overlap | Top-1 inv? |
|----------------|--------|---------------|------------|
| MaxCut         | 0.891  | 2/3           | Yes        |
| Circle Packing | 1.000  | 3/3           | No         |
| DiffBases      | 0.806  | 2/3           | No         |
| Flat Poly      | 1.000  | 3/3           | No         |
| TSP-100        | 1.000  | 3/3           | No         |
| Erdős          | 1.000  | 3/3           | No         |
| MolQED         | 0.952  | 3/3           | No         |
| TSP-50         | 1.000  | 3/3           | No         |

## 377 L Detailed MIG Analysis

### 378 L.1 Per-Task MIG Breakdown

Table 23: MIG per task for five configuration families (six optimization tasks with nonzero MIG; MaxCut and LJ-41 omitted as all configurations score zero). Bold marks Erdős sign flips.

| Family   | Cfg   | CPack | DBases | FPoly | TSP1  | Erdős        | MolQ  |
|----------|-------|-------|--------|-------|-------|--------------|-------|
| MAgICoRe | homo  | 0.00  | +0.05  | 0.00  | 0.00  | <b>+0.33</b> | −0.03 |
| MAgICoRe | mixed | 0.00  | −0.07  | 0.00  | −0.01 | <b>−0.28</b> | +0.09 |
| Debate   | homo  | −0.52 | +0.31  | +0.04 | −0.01 | +0.14        | +0.13 |
| Debate   | mixed | −0.60 | −0.14  | +0.02 | 0.00  | +0.14        | −0.04 |
| Chain    | homo  | 0.00  | 0.00   | 0.00  | −0.01 | <b>+0.43</b> | −0.01 |
| XChain   | mixed | −0.03 | −0.06  | −0.03 | +0.01 | <b>−0.08</b> | +0.01 |
| MoA      | homo  | −0.08 | +0.07  | 0.00  | −0.01 | +0.07        | +0.04 |
| MoA      | mixed | −0.24 | +0.15  | +0.06 | −0.01 | +0.07        | +0.10 |

### 379 L.2 Cross-Backbone Replication

Table 24: MAgICoRe vs. Best-of- $N$  on two optimization tasks, two backbones ( $n=15$  each). The loss replicates across backbones.

| Task      | Backbone | MAgICoRe | Best-of- $N$ | $d$   | $p$    |
|-----------|----------|----------|--------------|-------|--------|
| Flat Poly | Gemini   | 2.087    | 1.899        | +1.25 | 0.002  |
| Flat Poly | GPT-4o   | 2.951    | 2.251        | +1.30 | 0.001  |
| TSP-100   | Gemini   | 53,351   | 50,786       | +1.42 | 0.001  |
| TSP-100   | GPT-4o   | 53,529   | 50,128       | +2.64 | <0.001 |

### 380 L.3 Self-Refine Decomposition

Table 25: Self-Refine vs. MAgICoRe on constraint tasks. External critic (MAgICoRe) is the active ingredient; Self-Refine  $\approx$  BoN.

| Task                   | BoN | Self-Refine | MAgICoRe   | $p$ (MAgI vs BoN) |
|------------------------|-----|-------------|------------|-------------------|
| Circle Packing (feas.) | 27% | 27%         | <b>80%</b> | 0.005             |
| Knapsack (feas.)       | 0%  | 13%         | <b>47%</b> | 0.003             |
| 3AP-Free (feas.)       | 0%  | 7%          | <b>73%</b> | <0.001            |

381 On optimization tasks, MAgICoRe is significantly *worse* than BoN: TSP-100 ( $d=1.42$ ) and Flat  
382 Poly ( $d=1.25$ ). Iteration introduces noise rather than improvements.

### 383 L.4 MoA vs. MAgICoRe on Constraints

Table 26: Feasibility rates across constraint tasks. MoA-nosynth (pure diversity, no interaction) matches or beats MAgICoRe on 2/3 tasks.

| Task           | MAgICoRe   | BoN-Gem | MoA-nosynth | MoA-same | MoA-mixed |
|----------------|------------|---------|-------------|----------|-----------|
| Circle Packing | 80%        | 27%     | <b>100%</b> | 100%     | 100%      |
| Knapsack       | 47%        | 0%      | <b>70%</b>  | 60%      | 80%       |
| 3AP-Free       | <b>73%</b> | 0%      | 50%         | 100%     | 90%       |

384 On Circle Packing and Knapsack, pure diversity (MoA-nosynth) achieves comparable feasibility to  
385 multi-agent critique, without any interaction cost. On 3AP-Free (reasoning-required verification),  
386 MAgICoRe wins.

### 387 L.5 K-Curve / No Proxy Overfit

Table 27: K-curve on MaxCut: dev and hidden scores vs. number of evaluator calls. Both generally increase with  $K$ —no systematic proxy overfit.

| $K$ | Dev Mean | Hidden Mean | Calls Used |
|-----|----------|-------------|------------|
| 1   | 2413.6   | 3026.4      | 1.0        |
| 5   | 2443.8   | 3084.4      | 5.0        |
| 10  | 2468.0   | 3077.0      | 9.0        |
| 20  | 2474.2   | 3093.4      | 9.2        |
| 25  | 2442.4   | 3080.2      | 8.4        |

## 388 M Model Behavior Analysis

### 389 M.1 Strategy Classification

Table 28: Strategy frequency per backbone on Erdős and DiffBases. GPT-4o outputs a trivial constant on 100% of Erdős runs.

| Task      | Strategy           | Claude | GPT-4o     | Gemini     |
|-----------|--------------------|--------|------------|------------|
| Erdős     | Trivial constant   | 0/10   | <b>5/5</b> | <b>5/5</b> |
|           | Binary/block       | 4/10   | 0/5        | 0/5        |
|           | Mathematical       | 6/10   | 0/5        | 0/5        |
| DiffBases | Sidon sets         | 4/10   | 0/5        | 1/5        |
|           | Triangular nums    | 1/10   | <b>5/5</b> | 1/5        |
|           | Arith. progression | 1/10   | 0/5        | 2/5        |

## 390 M.2 Variance Decomposition

Table 29: Variance decomposition of single-shot Q-scores ( $n=5$  per backbone).  $B/W \geq 2$  indicates models are more different from each other than from themselves—a prerequisite for diversity to help.

| Task           | $\sigma_W^2$ | $\sigma_B^2$ | $B/W$ | Diversity useful? |
|----------------|--------------|--------------|-------|-------------------|
| Circle Packing | 0.017        | 0.253        | 15.2  | Yes               |
| MolQED         | 0.015        | 0.030        | 2.0   | Yes               |
| Flat Poly      | 0.054        | 0.070        | 1.3   | Marginal          |
| Erdős          | 0.050        | 0.061        | 1.2   | Marginal          |
| TSP-100        | 0.020        | 0.010        | 0.5   | No                |
| MaxCut         | 0.0001       | 0.00002      | 0.3   | No                |
| DiffBases      | 0.095        | 0.016        | 0.2   | No                |

## 391 M.3 Inter-Model Error Correlation

Table 30: Spearman rank correlations of per-seed hidden scores between model pairs. Low or negative  $\rho$  indicates uncorrelated errors.

| Task      | Cl-GPT $\rho$ | Cl-Gem $\rho$ | GPT-Gem $\rho$ | Error indep.    |
|-----------|---------------|---------------|----------------|-----------------|
| Erdős     | +0.18         | — (const)     | — (const)      | Yes             |
| DiffBases | +0.05         | +0.12         | +0.17          | Yes             |
| MolQED    | −0.45         | −0.20         | −0.33          | Yes (anti-corr) |
| TSP-100   | +0.38         | +0.50         | +0.41          | No (correlated) |

## 392 N Extended Experiment Results

### 393 N.1 Constraint Feasibility Gradient

Table 31: Constraint feasibility by configuration type (Gemini backbone,  $n=15$ ).

| Task         | S-Shot | BoN | VGS | MAGICoRe |
|--------------|--------|-----|-----|----------|
| Knapsack-50  | 7%     | 0%  | 27% | 47%      |
| 3AP-Free-100 | 0%     | 0%  | 13% | 73%      |

### 394 N.2 Crossover Full Results

Table 32: Crossover vs. MoA vs. MoA-nosynth: full hidden-score comparison ( $n=10$ ). Seeds matched. Seed-by-seed matches with MoA-nosynth measure whether crossover is merely copying the best proposer.

| Task         | Crossover     | MoA          | MoA-nosynth | Seed matches |
|--------------|---------------|--------------|-------------|--------------|
| DiffBases    | <b>10.99</b>  | 25.70        | 23.51       | 1/10         |
| Erdős        | 0.298         | <b>0.250</b> | 0.348       | 4/10         |
| Knapsack-50  | <b>842.2</b>  | 284.8        | 761.3       | 0/10         |
| Set-Cover-60 | <b>491</b>    | 100,343      | 200,309     | 0/10         |
| TSP-50       | <b>22,644</b> | 23,217       | 23,173      | 0/10         |

395 **Copying rates.** Crossover copied a proposer score in: DiffBases 20%, Erdős 40%, Knapsack 60%,  
 396 Set-Cover 50%, TSP 10%. Average:  $\sim 36\%$  vs.  $\sim 80\%$  for standard MoA synthesis.

397 **Feasibility.** Crossover achieves 100% feasibility on both Knapsack (vs. 70% MoA) and Set-Cover  
 398 (vs. 100% MoA).

### 399 N.3 Budget-2x Scaling

Table 33:  $2\times$  budget vs.  $1\times$  budget (hidden scores). 12/15 pairs improve with doubled budget, but gains are modest.

| Task      | Protocol    | $1\times$ | $2\times$ | $\Delta$ | Improved? |
|-----------|-------------|-----------|-----------|----------|-----------|
| DiffBases | BoN         | 15.37     | 12.88     | -16%     | Yes       |
| DiffBases | VGS         | 14.21     | 8.87      | -38%     | Yes       |
| DiffBases | Self-Refine | 16.98     | 18.99     | +12%     | No        |
| Erdős     | BoN         | 0.346     | 0.252     | -27%     | Yes       |
| Erdős     | MoA         | 0.408     | 0.267     | -34%     | Yes       |
| Erdős     | Self-Refine | 0.278     | 0.275     | -1%      | Yes       |
| MolQED    | BoN         | 0.761     | 0.806     | +6%      | Yes       |
| MolQED    | Self-Refine | 0.755     | 0.786     | +4%      | Yes       |
| MolQED    | VGS         | 0.772     | 0.788     | +2%      | Yes       |

### 400 N.4 Composable $2\times 2$ : Knapsack and Set-Cover

Table 34:  $2\times 2$  factorial on constraint tasks ( $n=10$ ). Diversity provides massive advantages on both tasks.

| Task         | Protocol          | devScore         | hiddenScore      |
|--------------|-------------------|------------------|------------------|
| Knapsack-50  | MoA (div + synth) | <b>720.6</b>     | <b>720.6</b>     |
|              | MoA-nosynth (div) | 662.4            | 449.4            |
|              | BoN (same)        | -626.2           | -626.2           |
|              | MoA-same          | -277.8           | -277.8           |
| Set-Cover-60 | MoA-nosynth (div) | <b>100,404.8</b> | <b>100,404.8</b> |
|              | MoA (div + synth) | 100,417.9        | 100,417.9        |
|              | BoN (same)        | 100,425.7        | 100,425.7        |
|              | MoA-same          | 439.2            | 439.2            |

401 Diversity advantage on Knapsack: +940 devScore. On Set-Cover: +99,966 devScore.

### 402 N.5 Prospective $2\times 2$ : Two New Tasks

Table 35: Prospective factorial on two new tasks ( $n=10$ ).

| Task           | Protocol          | devScore    | hiddenScore  |
|----------------|-------------------|-------------|--------------|
| Latin Square   | MoA-same          | <b>49.6</b> | <b>161.2</b> |
|                | MoA-nosynth (div) | 45.7        | 151.9        |
|                | MoA (div + synth) | 45.7        | 147.4        |
|                | BoN (same)        | 47.3        | 149.9        |
| Max-K-Coverage | MoA (div + synth) | <b>61.5</b> | <b>57.3</b>  |
|                | MoA-nosynth (div) | 60.8        | 54.9         |
|                | MoA-same          | 58.7        | 56.3         |
|                | BoN (same)        | 58.7        | 53.1         |

403 Same-model wins on Latin Square; diverse wins on Max-K-Coverage. The diversity advantage is  
404 task-dependent.

## 405 N.6 External 2×2: Six Additional Tasks

Table 36: External-2x2 results ( $n=5$ ). On harder tasks where many solutions are infeasible, diversity helps on 1/6 tasks but synthesis helps on 5/6.

| Task             | Diverse      | Same-model | Winner  |
|------------------|--------------|------------|---------|
| ext-knapsack-50  | <b>490.8</b> | −276.6     | Diverse |
| ext-maxsat-60    | 1015.2       | 1013.8     | ~Tie    |
| ext-mdtsp-40     | −2084.2      | −2057.4    | Same    |
| ext-mwis-40      | −156.4       | 202.4      | Same    |
| ext-partition-40 | −1291.6      | −624.4     | Same    |
| ext-tsp-25       | −1180.6      | −1145.4    | Same    |

406 Synthesis effect on external tasks: helps on 5/6 (average +214 devScore). On harder tasks with  
 407 frequent infeasibility, synthesis provides a correction mechanism—consistent with the verifiability  
 408 boundary (these tasks have checkable constraints).

## 409 N.7 Unreported Tasks

410 **HP-Folding.** Extremely hard task: 76/84 runs (90%) infeasible. Among the 8 valid folds, diverse  
 411 OSS configurations outperform same-model: `magicore-mixed-oss` achieves 4/5 valid folds versus  
 412 0/5 for same-model `MAGICoRe`.

413 **Set-Cover-60 (full-v2).** Filtering to feasible solutions: diverse MoA achieves the lowest cost (378.7)  
 414 versus 539–591 for same-model, while maintaining 90% feasibility. Quality–feasibility tradeoff  
 415 favors diversity.

416 **Graph Coloring.** Iterative methods (debate 40%, VGS 40%) perform worse than BoN  
 417 (80%). MoA (diverse) matches BoN at 80%. Without diverse debate/`MAGICoRe` variants, the  
 418 diversity×interaction effect cannot be tested.

## 419 O Prompts and Examples

### 420 O.1 Prompt Templates

421 Full prompt templates for all ten configurations are available in the artifact. Below we show two  
 422 representative structures.

#### 423 Proposer prompt (MoA, BoN, Single-Shot).

424 System: You are a mathematical optimization agent.  
 425 Your goal is to [OBJECTIVE]. Return ONLY executable  
 426 Python code that prints the solution as JSON.

427  
 428 User: [Full task description with parameters,  
 429 constraints, input data, and evaluation function.]

430  
 431 Output format: a single Python code block.

#### 432 Reviewer prompt (MAGICoRe, Debate).

433 System: You are a mathematical reviewer. Analyze  
 434 the proposed solution for correctness. List  
 435 specific errors, constraint violations, or  
 436 improvements.

437  
 438 User: Task: [OBJECTIVE]  
 439 Proposed solution (scored [VISIBLE\_SCORE]):  
 440 [SOLUTION\_CODE]  
 441 Provide numbered issues and concrete suggestions.

442 **O.2 Example Multi-Agent Exchange**

443 **Cross-Chain, Erdős, seed 1.** On Erdős, *lower* raw scores are better.

444 **Step 1 (Claude, dev = 0.500).** Claude reasons about minimizing overlap and produces a structured  
445 sparse function using binary patterns.

446 **Step 2 (GPT-4o, dev = 0.250).** GPT-4o sees Claude’s solution but outputs a trivial constant  $f = 0.5$ .  
447 GPT-4o defaults to near-constant functions on this task regardless of input context ( $Q=0.710$ ).

448 **Step 3 (Gemini, dev = 0.694).** Gemini produces a structured binary pattern but scores worse than  
449 GPT-4o. The chain’s visible-best selection picks GPT-4o’s output (0.250, lowest hence best).

450 The non-trivial solutions from Claude and Gemini are discarded in favor of a degenerate constant  
451 that contributes zero strategy diversity.